Aurora Theriault
Leo Dang
Marcus Hernandez
Quinlin Taylor

October 28, 2026

Specific Requirements

1. The game must be playable in the browser.
    1.1. The game will be compatible with Firefox and Chrome.
    1.2. The game will run at 20 frames per second or higher on most modern hardware.

2. The game must be accessible to people between 6th and 8th grade.
    2.1. Text will be in English and without errors in grammar or spelling.
        2.1.1. All fonts will be legible.

3. A question is a prompt to be answered by the player.
    3.1. Question topics will be sourced from the Massachusetts Department of Education Standards Mathematics Curriculum Framework.
        3.1.1. At least 10 topics will be used to create questions.
    3.2. Questions will have a singular, unambiguous answer.

4. A setting is any player-controllable value that can modify their playing experience.
    4.1. The game will have its volume as a setting.
    4.2. The game will have its keybinds as a setting.
    4.3. Question difficulty will be a setting.
    4.4. Settings will be persistent across sessions.

5. A map is composed of a path and towers.
    5.1. The path will have a defined start and end.
    5.2. Towers cannot be located on the path.
    5.3. Towers will have a number of upgradeable attributes.
        5.3.1. Attack damage may be an upgradeable attribute.
        5.3.2. Attack frequency may be an upgradeable attribute.
        5.3.3. Attack range may be an upgradeable attribute.
    5.4. The player will be able to view each tower's upgradeable attributes.
    5.5. There will be at least 3 maps.

6. A wave is a two-stage sequence that comprises the core gameplay loop.
    6.1. Each wave will begin with a rest stage.
        6.1.1. The player can choose when to end the rest stage.
        6.1.2. The rest stage will not have a time limit.
    6.2. Each wave will end with a defense stage.
        6.2.1. During the defense stage, enemies will spawn at the start of the map's path and progress to the end.
        6.2.2. The map's towers will attack and defeat the enemies.
            6.2.2.1. Enemies will visually communicate how many attacks remain for the player to defeat them.

6.2.3. The defense stage ends when all enemies have either been eliminated or progress off the screen.
6.2.4. Tower attributes can be upgraded during the defense stage by correctly answering a question.
6.2.4.1. The player will have one opportunity to answer each question.
6.2.4.2. Incorrectly answered questions will yield no bonus.
6.2.4.3. The player will be given the solution to incorrectly-answered questions.

7. A run is a sequence of waves.
7.1. Each wave will be more difficult than the previous.
7.1.1. The number of enemies may increase each wave.
7.1.2. The frequency of enemies may increase each wave.
7.1.3. The speed of the enemies may increase each wave.
7.1.4. The number of attacks needed to kill the enemies may increase each wave.
7.2. The player will be informed which wave they are in.
7.3. The player will be informed of the time spent in a run.
7.4. A run will end if 20 enemies reach the end of the path across all waves.
7.5. The player will be able to pause a run at any time.
7.5.1. While a run is paused, the time spent in the run will not increase.
7.5.2. The player will be able to change their settings while the run is paused.
7.5.3. The player will be able to exit a run while the run is paused.

8. The startup screen is the game's initial view after loading.
8.1. The game's title will be present on the startup screen.
8.2. The player will be able to start a new run from the startup screen.
8.3. The player will be able to change their settings from the startup screen.
8.4. The player will be able to view their best runs from the startup screen.
8.4.1. Runs will be scored by the furthest wave reached.
8.4.1.1. If two runs both reach the same furthest wave, the run with less playtime will be scored as better.
8.4.1.1.1. If two runs have the same furthest wave and the same playtime, then whichever run was achieved first will be scored as better.
8.4.2. The five best scoring runs will be viewable.
8.4.3. Best runs will be saved locally and persist through game closure.
8.5. The player will be able to view the game's credits and acknowledgements from the startup screen.

9. The game will be open source.